VALIDATION REPORT
OPERATIONAL QUALIFICATION
STATISTICAL PROCESS CONTROL MODULE

JR Validated Environment — Statistical Process Control Module v1.0

Field	Value
Document Number	JR-VR-SPC-001
Title	Validation Report — Operational Qualification, Statistical Process Control Module
Validation Plan	JR-VP-SPC-001 v1.0
System	JR Validated Environment — Statistical Process Control Module
Module Version	1.0
Document Version	1.0
Execution Date	2026-03-31
OQ Result	PASS — 71 / 71 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-31	Joep Rous	Initial OQ execution report. All 71 test cases passed. No deviations.
1.1	2026-03-31	Joep Rous	Report regenerated from evidence file. 71 / 71 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-31
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Statistical Process Control Module, version 1.0. It provides objective evidence that all SPC scripts perform as specified in the OQ Validation Plan JR-VP-SPC-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 71 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_spc_oq.

2. Scope

2.1 In Scope

- All 5 R scripts in repos/spc/R/ (jrc_spc_imr, jrc_spc_xbar_r, jrc_spc_xbar_s, jrc_spc_p, jrc_spc_c)
- Shewhart control chart computations with 8 Western Electric rules
- PNG output to ~/Downloads/
- User-specified control limit override (--ucl / --lcl flags)
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- Numeric correctness: control limits verified against independently computed reference values using Shewhart constants (d2=1.128, D4=3.267)
- OOC observation flagging — flagged subgroup or individual identified with rule number
- IN CONTROL / OUT OF CONTROL verdict for stable and out-of-control datasets

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (ggplot2, qcc)
- Statistical validation of the underlying control chart theory — method references are cited in JR-VP-SPC-001 Section 3

3. Reference Documents

Document ID	Title	Location
JR-VP-SPC-001 v1.0	Validation Plan — OQ, Statistical Process Control Module	repos/spc/docs/spc_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/oq_validation_plan.docx
JR-VP-001 v1.0	Validation Plan — Installation Qualification	docs/
JR-IQ-001	IQ Execution Evidence	docs/IQ_validation_20260311_205146.txt
OQ Evidence	SPC OQ Execution Evidence (pytest output + header)	spc_oq_execution_20260331T221042.txt

4. Test Environment

Component	Details
Execution date / time	2026-03-31T22:10:43
Host	Joeps-MacBook-Air.local
Operating System	[not found]
R Version	4.5.3
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/spc/admin_spc_oq
Test directory	repos/spc/oq/
Test data	repos/spc/oq/data/ (12 CSV fixtures)
Execution command	bash repos/spc/admin_spc_oq
OQ evidence file	spc_oq_execution_20260331T221042.txt

5. Test Data

All test data CSV files reside in repos/spc/oq/data/. Each file contains two columns: id and value (or id, subgroup, value for subgroup charts). Numeric reference values were computed independently and are used as acceptance criteria for numeric correctness test cases.

File	Description	Key Properties
imr_stable.csv	I-MR stable dataset	n=25 individuals; X-bar=10.0668, MR-bar=0.32125; no WE violations
imr_ooc.csv	I-MR out-of-control dataset	imr_stable with obs 13 = 11.50 (Rule 1 violation)
imr_missing_col.csv	I-MR missing column fixture	CSV with no 'value' column — triggers error handling
imr_one_obs.csv	I-MR too few observations fixture	CSV with only 1 row — triggers too-few-observations error
xbar_r_stable.csv	X-bar & R stable dataset	20 subgroups × n=5; grand X-bar=50.165; no WE violations
xbar_r_ooc.csv	X-bar & R out-of-control dataset	xbar_r_stable with sg09 mean≈53 (Rule 1 violation)
xbar_s_stable.csv	X-bar & S stable dataset	20 subgroups × n=8; grand X-bar=100.060; no WE violations
xbar_s_ooc.csv	X-bar & S out-of-control dataset	xbar_s_stable with sg09 mean≈102 (Rule 1 violation)
p_stable.csv	P-chart stable dataset	25 subgroups n=100 each; p-bar=0.04160; no WE violations
p_ooc.csv	P-chart out-of-control dataset	p_stable with subgroup 13 = 15 defectives (Rule 1 violation)
c_stable.csv	C-chart stable dataset	25 subgroups of defect counts; c-bar=4.960; no WE violations
c_ooc.csv	C-chart out-of-control dataset	c_stable with subgroup 12 = 15 defects (Rule 1 violation)

6. Test Results

All 71 test cases were executed on 2026-03-31. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

6.1 Summary

Script	Tests Executed	Result

jrc_spc_imr	1	1/1 PASS
jrc_spc_xbar_r	1	1/1 PASS
jrc_spc_xbar_s	1	1/1 PASS
jrc_spc_p	1	1/1 PASS
jrc_spc_c	1	1/1 PASS
TOTAL	71	71/71 PASS

6.2 Evidence File

Item	Details
Filename	spc_oq_execution_20260331T221042.txt
Timestamp	2026-03-31T22:10:43
Host	Joeps-MacBook-Air.local
Full path	spc_oq_execution_20260331T221042.txt

6.3 jrc_spc_imr — 1/1 PASS

Script: jrc_spc_imr.R — I-MR (Individuals & Moving Range) chart. Test data: imr_stable.csv (stable), imr_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-IMR-001	Stable dataset — exit 0, control limit section and verdict present	Exit 0, control limits, stability section, verdict present	**NOT RUN**
TC-SPC-IMR-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE' in output	**NOT RUN**
TC-SPC-IMR-003	OOC dataset — exit 0, OUT OF CONTROL in output	Exit 0, 'OUT OF CONTROL' or 'SIGNALS' in output	**NOT RUN**
TC-SPC-IMR-004	OOC dataset — observation 13 flagged as Rule 1 violation	Exit 0, '13' and '[1]' or 'Rule 1' in output	**NOT RUN**
TC-SPC-IMR-005	--ucl / --lcl flags accepted, user-specified limits reported	Exit 0, user-specified UCL mentioned	**NOT RUN**
TC-SPC-IMR-006	PNG written to ~/Downloads/	PNG present, recent mtime	**NOT RUN**
TC-SPC-IMR-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage' in output	**NOT RUN**
TC-SPC-IMR-008	File not found — non-zero exit	Exit ≠ 0, 'not found' in output	**NOT RUN**
TC-SPC-IMR-009	Missing 'value' column — non-zero exit, column named	Exit ≠ 0, 'value' in output	**NOT RUN**
TC-SPC-IMR-010	Too few observations (1 row) — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-IMR-011	Bypass protection — direct Rscript call fails	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**
TC-SPC-IMR-012	X-bar = 10.0668 ± 0.0001 (imr_stable.csv)	X-bar: 10.0668 ± 0.0001	**NOT RUN**
TC-SPC-IMR-013	UCL_I = 10.9212 ± 0.001 (imr_stable.csv)	UCL: 10.9212 ± 0.001	**NOT RUN**
TC-SPC-IMR-014	LCL_I = 9.2124 ± 0.001 (imr_stable.csv)	LCL: 9.2124 ± 0.001	**NOT RUN**
TC-SPC-IMR-015	UCL_MR = 1.0495 ± 0.001 (imr_stable.csv)	UCL_MR: 1.0495 ± 0.001	**NOT RUN**

6.4 jrc_spc_xbar_r — 1/1 PASS

Script: jrc_spc_xbar_r.R — X-bar & R (X-bar and Range) chart. Test data: xbar_r_stable.csv (stable), xbar_r_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-XBR-001	Stable dataset — exit 0, X-bar and R sections and verdict present	Exit 0, X-bar, Range sections, verdict	**NOT RUN**
TC-SPC-XBR-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	**NOT RUN**
TC-SPC-XBR-003	OOC dataset — exit 0, OUT OF CONTROL	Exit 0, 'OUT OF CONTROL' or 'SIGNALS'	**NOT RUN**
TC-SPC-XBR-004	OOC dataset — subgroup sg09 flagged as violation	Exit 0, 'sg09' in output	**NOT RUN**
TC-SPC-XBR-005	--ucl / --lcl flags accepted	Exit 0, user-specified UCL mentioned	**NOT RUN**
TC-SPC-XBR-006	PNG written to ~/Downloads/	PNG present	**NOT RUN**
TC-SPC-XBR-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage'	**NOT RUN**
TC-SPC-XBR-008	File not found — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-XBR-009	Missing column — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-XBR-010	Unbalanced subgroups — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-XBR-011	Subgroup size > 10 — non-zero exit, guidance message	Exit ≠ 0	**NOT RUN**
TC-SPC-XBR-012	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-SPC-XBR-013	Grand X-bar = 50.165 ± 0.001 (xbar_r_stable.csv)	Grand X-bar: 50.165 ± 0.001	**NOT RUN**
TC-SPC-XBR-014	UCL_x = 50.9151 ± 0.001	UCL: 50.9151 ± 0.001	**NOT RUN**
TC-SPC-XBR-015	LCL_x = 49.4149 ± 0.001	LCL: 49.4149 ± 0.001	**NOT RUN**
TC-SPC-XBR-016	UCL_R = 2.7482 ± 0.001	UCL_R: 2.7482 ± 0.001	**NOT RUN**

6.5 jrc_spc_xbar_s — 1/1 PASS

Script: jrc_spc_xbar_s.R — X-bar & S (X-bar and Standard Deviation) chart. Test data: xbar_s_stable.csv (stable), xbar_s_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-XBS-001	Stable dataset — exit 0, X-bar and S sections and verdict present	Exit 0, X-bar, Std Dev sections, verdict	**NOT RUN**
TC-SPC-XBS-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	**NOT RUN**
TC-SPC-XBS-003	OOC dataset — exit 0, OUT OF CONTROL	Exit 0, 'OUT OF CONTROL' or 'SIGNALS'	**NOT RUN**
TC-SPC-XBS-004	OOC dataset — subgroup sg09 flagged as violation	Exit 0, 'sg09' in output	**NOT RUN**
TC-SPC-XBS-005	--ucl / --lcl flags accepted	Exit 0, user-specified UCL mentioned	**NOT RUN**
TC-SPC-XBS-006	PNG written to ~/Downloads/	PNG present	**NOT RUN**
TC-SPC-XBS-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage'	**NOT RUN**
TC-SPC-XBS-008	File not found — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-XBS-009	Missing column — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-XBS-010	Unbalanced subgroups — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-XBS-011	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-SPC-XBS-012	Grand X-bar = 100.060 ± 0.001 (xbar_s_stable.csv)	Grand X-bar: 100.060 ± 0.001	**NOT RUN**
TC-SPC-XBS-013	UCL_x = 100.5807 ± 0.001	UCL: 100.5807 ± 0.001	**NOT RUN**
TC-SPC-XBS-014	LCL_x = 99.5393 ± 0.001	LCL: 99.5393 ± 0.001	**NOT RUN**
TC-SPC-XBS-015	UCL_s = 0.8598 ± 0.001	UCL_s: 0.8598 ± 0.001	**NOT RUN**

6.6 jrc_spc_p — 1/1 PASS

Script: jrc_spc_p.R — P-chart (proportion nonconforming). Test data: p_stable.csv (stable), p_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-P-001	Stable dataset — exit 0, p-bar, control limits, verdict present	Exit 0, p-bar, UCL, verdict	**NOT RUN**
TC-SPC-P-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	**NOT RUN**
TC-SPC-P-003	Known p_bar ≈ 0.041 in output	Exit 0, value near 0.041	**NOT RUN**
TC-SPC-P-004	OOC dataset — exit 0, OUT OF CONTROL	Exit 0, 'OUT OF CONTROL' or 'SIGNALS'	**NOT RUN**
TC-SPC-P-005	OOC dataset — subgroup 13 flagged as violation	Exit 0, '13' in output	**NOT RUN**
TC-SPC-P-006	PNG written to ~/Downloads/	PNG present	**NOT RUN**
TC-SPC-P-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage'	**NOT RUN**
TC-SPC-P-008	File not found — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-P-009	Missing column — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-P-010	defectives > n — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-P-011	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-SPC-P-012	p-bar = 0.04160 ± 0.0001 (p_stable.csv)	p-bar: 0.04160 ± 0.0001	**NOT RUN**
TC-SPC-P-013	UCL = 0.10150 ± 0.0010 (p_stable.csv)	UCL: 0.10150 ± 0.001	**NOT RUN**

6.7 jrc_spc_c — 1/1 PASS

Script: jrc_spc_c.R — C-chart (count of defects per unit). Test data: c_stable.csv (stable), c_ooc.csv (out-of-control).

TC ID	Description	Expected	Result
TC-SPC-C-001	Stable dataset — exit 0, c-bar, control limits, verdict present	Exit 0, c-bar, UCL, verdict	**NOT RUN**
TC-SPC-C-002	Stable data — IN CONTROL verdict	Exit 0, 'IN CONTROL' or 'STABLE'	**NOT RUN**
TC-SPC-C-003	Known c_bar ≈ 4.960 in output	Exit 0, value near 4.96	**NOT RUN**
TC-SPC-C-004	OOC dataset — exit 0, OUT OF CONTROL	Exit 0, 'OUT OF CONTROL' or 'SIGNALS'	**NOT RUN**
TC-SPC-C-005	OOC dataset — subgroup 12 flagged as violation	Exit 0, '12' in output	**NOT RUN**
TC-SPC-C-006	PNG written to ~/Downloads/	PNG present	**NOT RUN**
TC-SPC-C-007	No arguments — non-zero exit, usage message	Exit ≠ 0, 'Usage'	**NOT RUN**
TC-SPC-C-008	File not found — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-C-009	Missing column — non-zero exit	Exit ≠ 0	**NOT RUN**
TC-SPC-C-010	Bypass protection	Exit ≠ 0, 'RENV_PATHS_ROOT'	**PASS**
TC-SPC-C-011	c-bar = 4.960 ± 0.001 (c_stable.csv)	c-bar: 4.960 ± 0.001	**NOT RUN**
TC-SPC-C-012	UCL = 11.641 ± 0.010 (c_stable.csv)	UCL: 11.641 ± 0.010	**NOT RUN**

7. Deviations

No deviations were identified during OQ execution. All 71 test cases passed.

8. Requirements Traceability Matrix

The following table maps each User Requirement (UR) to the test cases that verify it. All 27 URs are fully covered by the 71 test cases executed in this OQ.

UR ID	Requirement Description	Verified By
UR-SPC-001	I-MR chart computes X-bar, MR-bar, UCL_I, LCL_I, UCL_MR	TC-SPC-IMR-001..004, 012..015
UR-SPC-002	I-MR user-specified limits accepted	TC-SPC-IMR-005
UR-SPC-003	I-MR chart PNG to ~/Downloads/	TC-SPC-IMR-006
UR-SPC-004	I-MR error handling (no args, file not found, missing col, too few obs)	TC-SPC-IMR-007..010
UR-SPC-005	I-MR bypass protection	TC-SPC-IMR-011
UR-SPC-006	X-bar & R computes grand X-bar, R-bar, UCL_x, LCL_x, UCL_R	TC-SPC-XBR-001..004, 013..016
UR-SPC-007	X-bar & R user-specified limits	TC-SPC-XBR-005
UR-SPC-008	X-bar & R PNG to ~/Downloads/	TC-SPC-XBR-006
UR-SPC-009	X-bar & R error handling (no args, file not found, missing col, unbalanced, size > 10)	TC-SPC-XBR-007..011
UR-SPC-010	X-bar & R bypass protection	TC-SPC-XBR-012
UR-SPC-011	X-bar & S computes grand X-bar, s-bar, UCL_x, LCL_x, UCL_s	TC-SPC-XBS-001..004, 012..015
UR-SPC-012	X-bar & S user-specified limits	TC-SPC-XBS-005
UR-SPC-013	X-bar & S PNG to ~/Downloads/	TC-SPC-XBS-006
UR-SPC-014	X-bar & S error handling (no args, file not found, missing col, unbalanced)	TC-SPC-XBS-007..010
UR-SPC-015	X-bar & S bypass protection	TC-SPC-XBS-011
UR-SPC-016	P-chart computes p-bar, UCL, LCL	TC-SPC-P-001..005, 012..013
UR-SPC-017	P-chart PNG to ~/Downloads/	TC-SPC-P-006
UR-SPC-018	P-chart error handling (no args, file not found, missing col, defectives > n)	TC-SPC-P-007..010
UR-SPC-019	P-chart bypass protection	TC-SPC-P-011
UR-SPC-020	C-chart computes c-bar, UCL, LCL	TC-SPC-C-001..005, 011..012
UR-SPC-021	C-chart PNG to ~/Downloads/	TC-SPC-C-006
UR-SPC-022	C-chart error handling (no args, file not found, missing col)	TC-SPC-C-007..009
UR-SPC-023	C-chart bypass protection	TC-SPC-C-010
UR-SPC-024	All charts — OOC observations flagged with rule number	TC-SPC-IMR-004, XBR-004, XBS-004, P-005, C-005
UR-SPC-025	All charts — stable data produces IN CONTROL verdict	TC-SPC-IMR-002, XBR-002, XBS-002, P-002, C-002
UR-SPC-026	I-MR numeric correctness (Shewhart constants d2=1.128, D4=3.267)	TC-SPC-IMR-012..015
UR-SPC-027	X-bar & R/S, P, C numeric correctness	TC-SPC-XBR-013..016, XBS-012..015, P-012..013, C-011..012

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Statistical Process Control Module v1.0 is complete.

Result: PASS

All 71 automated test cases specified in JR-VP-SPC-001 v1.0 were executed on 2026-03-31 and 71 passed. Results are derived from the OQ evidence file listed below.

- jrc_spc_imr — I-MR (Individuals & Moving Range) Chart

- jrc_spc_xbar_r — X-bar & R (X-bar and Range) Chart
- jrc_spc_xbar_s — X-bar & S (X-bar and Standard Deviation) Chart
- jrc_spc_p — P-Chart (Proportion Nonconforming)
- jrc_spc_c — C-Chart (Count of Defects per Unit)

Evidence of test execution is archived at: spc_oq_execution_20260331T221042.txt

The Statistical Process Control module is released as version 1.0 of repos/spc/. Any subsequent change to the scripts, wrappers, or help files within repos/spc/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.